Project: Worked examples
Title: Pipe expansion, loops and anchor forces (screening)
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## 1. Scope and method

Screening only — not a piping stress analysis. Thermal growth of a straight run between anchors, the fully restrained stress and force, the leg an offset leg or a U-loop (two guided cantilevers) needs to absorb the growth within the owner's allowable displacement-stress range Sa and whether the available leg is long enough (an offset leg or Z-bend also loads its run or runs with the leg's own growth), the force on the anchors and the largest root moment of the guided-cantilever model, the screening ratio $D \cdot Y/(L - U)^2$ against the owner's K1 (PASS, or the line needs a formal stress analysis), and, for an axial bellows, the anchor force $P \cdot A_{eff}$ plus its spring force. No allowable stress, K1 or code table ships.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Pipe section | $d_i = OD - 2 \cdot t$; $A_m = \pi/4 \cdot (OD^2 - d_i^2)$; $I = \pi/64 \cdot (OD^4 - d_i^4)$; $Z = I/(OD/2)$ |
| Temperature change | $\Delta T = T_{operate} - T_{install}$, or as entered; the movement uses $\lvert\Delta T\rvert$ and its sign gives the direction |
| Run that grows | $L_{tot} = L$, or $L + L2$ for a Z-bend; thermal growth $\Delta L = \alpha \cdot L_{tot} \cdot \lvert\Delta T\rvert$ |
| Fully restrained pipe | $\sigma_r = E \cdot \alpha \cdot \lvert\Delta T\rvert$ (E at the installation temperature); $F_r = E_r \cdot \alpha \cdot \lvert\Delta T\rvert \cdot A_m$ ($E_r = E_{hot}$, else E) |
| Movement one leg must absorb | $\Delta = \Delta L$ for an offset leg or Z-bend; $\Delta = \Delta L/2$ for a U-loop (two identical legs in series, the cross leg a rigid guide) |
| Guided cantilever, required leg | $L_{req} = \sqrt{(3 \cdot E \cdot OD \cdot \Delta/Sa)}$, the leg at which the bending stress M/Z reaches Sa; PASS when $L_{req} \le$ the available leg H |
| Force and moment of the available leg | $F_{leg} = 12 \cdot E_r \cdot I \cdot \Delta/H^3$ (acts along the run); $M_{leg} = 6 \cdot E_r \cdot I \cdot \Delta/H^2$ ($= F_{leg} \cdot H/2$); $\sigma_{leg} = 3 \cdot E \cdot OD \cdot \Delta/H^2 = Sa \cdot (L_{req}/H)^2$ |
| Offset leg or Z-bend, the runs bend for the leg's growth $\Delta_{leg} = \alpha \cdot \lvert\Delta T\rvert \cdot H$ | the runs act in series under one force $F_{run} = 12 \cdot E_r \cdot I \cdot \Delta_{leg}/(L^3 + L2^3)$ ($L2 = 0$ for an L-bend) acting along the leg; run i takes the share $L_i^3/(L^3 + L2^3)$ of $\Delta_{leg}$, with root stress $\sigma_i = 3 \cdot E \cdot OD \cdot \Delta_i/L_i^2$ and moment $M_i = F_{run} \cdot L_i/2$; each run is checked as its own cantilever, PASS when $\sqrt{(\sigma_i/Sa)} \le 1$ (for an L-bend the run needed is $L_{req,run} = \sqrt{(3 \cdot E \cdot OD \cdot \Delta_{leg}/Sa)}$) |
| Force on the anchors | the two components are perpendicular, $F_{leg}$ along the run and $F_{run}$ along the leg, so the force is $\sqrt{(F_{leg}^2 + F_{run}^2)}$; the moment reported is the largest root moment of the legs and runs in the guided-cantilever model |
| Flexibility screening | $OD \cdot Y/(L_{dev} - U)^2 \le K1$ with $Y = \alpha \cdot \lvert\Delta T\rvert \cdot U$, the free growth between the anchors; offset leg or Z-bend $L_{dev} = L_{tot} + H$, $U = \sqrt{(L_{tot}^2 + H^2)}$; U-loop $L_{dev} = L + 2 \cdot H$, $U = L$ (so $Y = \Delta L$); PASS, or the line needs a formal stress analysis |

| WHAT IT CALCULATES | RELATION |
|---|---|
| Axial bellows (no leg or loop, no leg check, no screening) | anchor force = $P \cdot A_{eff} + k \cdot \Delta L$; the movement is checked against the bellows' rated axial movement when given |

Basis: Process piping — flexibility analysis, exemption from formal analysis[1](para. 319.4.1 (screening relation; number to be checked against the owner's copy)); Process piping — displacement stress range and reactions[1](para. 319.3.2 and 319.5 (reference and hot moduli; numbers to be checked against the owner's copy)); Metallic industrial piping — flexibility analysis[2](cl. 12 (number to be checked against the owner's copy)).

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| E | Young's modulus of the pipe material at the installation (reference) temperature, for the stress check, in GPa (steel 210, copper 120, PE and PP about 1; not MPa) | 120 | GPa | assumption: W3 seed 'Copper (generic)', indicative; replace with the project's material record. | 0.5 ... 500 GPa |
| α | Mean coefficient of thermal expansion between the install and operating temperatures | 17 | um/(m·K) | assumption: W3 seed 'Copper (generic)' (17e-6 1/K), indicative; replace with the project's material record. | 1 ... 250 um/(m·K) |
| OD | Pipe outside diameter | 54 | mm | drawing: DRG-M-212 rev B | 10 ... 1500 mm |
| $t_{wall}$ | Pipe wall thickness | 1.5 | mm | drawing: DRG-M-212 rev B | 0.5 ... 60 mm |
| L | Anchor spacing: the straight length that grows (an offset leg or Z-bend: the first run, which also bends for the leg's growth) | 24 | m | drawing: DRG-M-212 rev B | 1 ... 500 m |
| $temp_{mode}$ | Temperature change from | install and operating temperatures | | assumption: Install and operating temperatures are both known. | any |
| $T_{install}$ | Temperature at installation (when the run is anchored) | 10 | °C | assumption: Mean site temperature when the riser is anchored. | -50 ... 50 °C |
| $T_{operate}$ | Operating temperature (hot or cold) | 70 | °C | spec: SPEC-M-04 §3: heating flow temperature | -50 ... 500 °C |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| layout | What absorbs the movement | axial bellows (no leg or loop: the bellows absorbs the movement) | | drawing: DRG-M-212 rev B | any |
| $P_b$ | Axial bellows governing pressure, gauge, at or above zero: the design or test pressure that governs the anchor sizing (a vacuum bellows is outside this sheet; compression and extension ratings can differ, enter the governing one) | 6 | bar | spec: Heating circuit design pressure, gauge | 0.1 … 40 bar |
| $A_{eff}$ | Axial bellows effective area $A_{eff}$ (manufacturer's data) | 40 | cm² | assumption: Placeholder effective area; replace with the manufacturer's datasheet value. | 1 … 5000 cm² |
| $k_b$ | Axial bellows spring rate (manufacturer's data; blank = no spring force) | 80 | N/mm | assumption: Placeholder spring rate; replace with the manufacturer's datasheet value. | 1 … 5000 N/mm |
| $dL_{rated}$ | Rated axial movement of the bellows (manufacturer's data; blank = movement not checked) | 40 | mm | assumption: Placeholder rated axial movement; replace with the manufacturer's datasheet value. | 1 … 500 mm |

## 3. Assumptions

A1. Screening only — not a piping stress analysis. The result tells whether a simple leg or loop is plausible and what it puts on its anchors; it does not replace a flexibility analysis of the real routing (elbow flexibility, intensification, torsion, branches, restraint friction, supports, sustained and occasional loads). (Stated scope.)
A2. A guided cantilever: the available leg is straight, uniform, fixed against rotation at both ends and loaded only by the movement it absorbs (stiffness $12 \cdot E \cdot I / H^3$, end moment $F \cdot H / 2$). Elbows are taken as rigid and the leg as slender: the beam model is not valid for legs shorter than about ten diameters. (Beam theory.)
A3. No stress-intensification factor and no elbow flexibility are applied: the stress at the elbow root of a real bend is higher than the beam stress M/Z, by the intensification factor of the code, so a leg that passes here can still exceed Sa at the elbow. This is the main reason the result is a screen and not a stress analysis. (Stated limit.)
A4. A U-loop is two identical legs in series, each a guided cantilever taking half the growth, the cross leg a rigid guide; the force along the line is the same F in both legs. The moment M is the moment at the leg

roots (the corners of the loop); it reaches an anchor only where the leg ends at the anchor (an offset leg). The growth of the loop's own legs, which the cross leg has to absorb, is not checked. (Loop model.)
A5. An offset leg is anchored at both ends: the run L bends for the growth of the leg ($\alpha \cdot |\Delta T| \cdot H$) as a second guided cantilever and is checked against the same Sa. A Z-bend (second run L2) has the two runs in series under one force, each taking its share of the leg's growth in proportion to its flexibility, $L_i^3$, and each checked as its own cantilever; the leg takes the growth of both runs. The force on the anchors is the resultant of the force along the run (the leg's) and the force along the leg (the runs'), which are perpendicular. (Offset-leg and Z-bend model.)
A6. $anchor_{moment}$ is the largest root moment of the legs and runs in the guided-cantilever simplification ($M = F \cdot L/2$ with each end fixed against rotation). It is not the moment on a frame anchor, which depends on the whole frame (the stiffness of the other legs, the elbow rotations, the anchor's own fixity); a formal analysis is needed before an anchor is sized for moment. (Stated limit.)
A7. The material properties are single values: α the mean coefficient between the install and operating temperatures; E the modulus at the installation (reference) temperature, used for the stress check, as the piping codes do (ASME B31.3 para. 319.3.2 and 319.5, numbers to be checked against the owner's copy); $E_{hot}$, when given, is used only for the reactions on the anchors. Take them from a W3 material applied at the right temperature. (Material library; code convention to be checked.)
A8. Sa and K1 are the owner's records from the owner's own copy of the piping code (or the design basis), with their provenance; no allowable stress, K1 or code constant ships. The sheet applies the relation to the numbers entered and does not know which code, material or load case they belong to. (No table from a standard is shipped.)
A9. No friction, weight, spring-hanger or cold-spring effect is included in the leg force; the anchor force adds the friction of the sliding supports, $\mu \cdot w \cdot L$, only when μ and the weight per metre are entered (a warning says so when they are not). The axial bellows anchor force is the pressure thrust $P \cdot A_{eff}$ plus the spring force $k \cdot \Delta L$ at the governing pressure entered (the design or the test pressure, whichever governs; a bellows' compression and extension ratings can differ): guide friction and any other bellows load come from the manufacturer's data. (Stated limit.)
A10. E (Young's modulus of the pipe material at the installation (reference) temperature, for the stress check, in GPa (steel 210, copper 120, PE and PP about 1; not MPa)) is taken as 120 GPa. (W3 seed 'Copper (generic)', indicative; replace with the project's material record.)
A11. α (Mean coefficient of thermal expansion between the install and operating temperatures) is taken as 17 um/(m*K). (W3 seed 'Copper (generic)' (17e-6 1/K), indicative; replace with the project's material record.)
A12. $temp_{mode}$ (Temperature change from) is taken as install and operating temperatures. (Install and operating temperatures are both known.)
A13. $T_{install}$ (Temperature at installation (when the run is anchored)) is taken as 10 degC. (Mean site temperature when the riser is anchored.)
A14. $A_{eff}$ (Axial bellows effective area $A_{eff}$ (manufacturer's data)) is taken as 40 cm^2. (Placeholder effective area; replace with the manufacturer's datasheet value.)
A15. $k_b$ (Axial bellows spring rate (manufacturer's data; blank = no spring force)) is taken as 80 N/mm. (Placeholder spring rate; replace with the manufacturer's datasheet value.)
A16. $dL_{rated}$ (Rated axial movement of the bellows (manufacturer's data; blank = movement not checked)) is taken as 40 mm. (Placeholder rated axial movement; replace with the manufacturer's datasheet value.)
A17. Layout: an axial bellows absorbs the whole movement of 24.48 mm; there is no leg check and no screening. The anchors take the pressure thrust $P \cdot A_{eff}$ plus the spring force $k \cdot \Delta L$; guide friction, lateral and angular movement and the bellows' own limits are the manufacturer's. Screening only — not a piping stress analysis. (Bellows layout.)

## 4. Working

[9] $dT = T_{operate} - T_{install}$
= 70 degC − 10 degC
= 60 K
Temperature change, operating − install
[11] $E_r = E$
= 120 GPa
= 120 GPa
Modulus for the reactions (E, no $E_{hot}$ given)
[12] $D_i = OD - 2 * t_{wall}$
= 54 mm − 2 × 1.5 mm
= 51 mm
Inside diameter
[13] $A_m = pi / 4 * (OD^2 - D_i^2)$
= π / 4 × ((54 mm)^2 − (51 mm)^2)
= 247.4 mm²
Metal cross-section
[14] $I_x = pi / 64 * (OD^4 - D_i^4)$
= π / 64 × ((54 mm)^4 − (51 mm)^4)
= 8.5307 cm⁴
Second moment of area of the pipe section
[15] $Z_x = I_x / (OD / 2)$
= 8.5307 cm⁴ / (54 mm / 2)
= 3.1595 cm³
Section modulus
[16] $L_{tot} = L$
= 24 m
= 24 m
Length that grows between the anchors
[17] $dL = \alpha * L_{tot} * abs(dT)$
= 17 um/(m·K) × 24 m × abs(60 K)
= 24.48 mm
Thermal growth between the anchors, as a magnitude
[18] $dL_{signed} = \alpha * L_{tot} * dT$
= 17 um/(m·K) × 24 m × 60 K
= 24.48 mm
Thermal movement with its sign: positive lengthens, negative shortens
[19] $\sigma_{res} = E * \alpha * abs(dT)$
= 120 GPa × 17 um/(m·K) × abs(60 K)
= 122.4 MPa
Stress in the fully restrained pipe (E at the installation temperature)
[20] $F_{res} = E_r * \alpha * abs(dT) * A_m$
= 120 GPa × 17 um/(m·K) × abs(60 K) × 247.4 mm²
= 30.282 kN
Force on the anchors of the fully restrained pipe
[21] $F_{friction}$ = support friction not entered
= μ and $w_{pipe}$ blank
= 0 N
Friction of the sliding supports (not entered: none in the anchor force)
[24] $F_{thrust} = P_b * A_{eff}$
= 6 bar × 40 cm²
= 2.4 kN
Bellows pressure thrust on the anchors, at the governing (design or test) pressure

[26] $F_{spring} = k_b * dL$
= 80 N/mm × 24.48 mm
= 1958.4 N
Bellows spring force at the full movement
[27] $anchor_{force} = F_{thrust} + F_{spring} + F_{friction}$
= 2.4 kN + 1958.4 N + 0 N
= 4358.4 N
Force on the anchors: pressure thrust plus spring force (plus support friction if μ and $w_{pipe}$ are entered)
[28] $anchor_{moment}$ = axial bellows: no bending
= the movement is axial
= 0 N·m
Moment at the anchors: none from an axial bellows
[30] $u_{bellows} = dL / dL_{rated}$
= 24.48 mm / 40 mm
= 0.612
Movement over the bellows' rated axial movement
[31] pass = the rated movement is given and met
= 1 check(s)
= 1
1 when the movement is within the bellows' rated axial movement; 0 when it is not or when no rating was entered

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $D_i$ | 51 mm | Inside diameter |
| $A_m$ | 247.4 mm² | Metal cross-section |
| $I_x$ | 8.5307 cm⁴ | Second moment of area |
| $Z_x$ | 3.1595 cm³ | Section modulus |
| dT | 60 K | Temperature change |
| $L_{tot}$ | 24 m | Length that grows |
| dL | 24.48 mm | Thermal movement between the anchors (magnitude) |
| $dL_{signed}$ | 24.48 mm | Thermal movement, signed |
| $\sigma_{res}$ | 122.4 MPa | Fully restrained stress |
| $F_{res}$ | 30.282 kN | Fully restrained force |
| direction | "growth: the pipe lengthens; fully restrained it presses on its anchors (compression)" | Direction of the movement |
| $F_{friction}$ | 0 N | Support friction on the run |
| $F_{thrust}$ | 2.4 kN | Bellows pressure thrust |
| $anchor_{force}$ | 4358.4 N | Force on the anchors |
| $anchor_{moment}$ | 0 N·m | Moment at the anchors |
| $F_{spring}$ | 1958.4 N | Bellows spring force |
| $u_{bellows}$ | 0.612 | Movement over the rated movement |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| screen | "not applicable: the bellows absorbs the movement" | Screening result |
| pass | 1 | All checks pass |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Movement against the rated axial movement of the bellows (40 mm) | $u_{bellows}$ = 0.612 | $u_{bellows}$ ≤ 1 | 0.612 | |

## 7. Applicability

WARNING — record: Tabulated values not taken from a reference record: E, α (Material properties); OD, $t_{wall}$ (Pipe outside diameter and wall); $A_{eff}$, $k_b$, $dL_{rated}$ (Axial bellows data). Enter each as one so its source, edition and licence travel with it.
WARNING — friction: μ: the anchor force holds no support friction. Every slide or guide between the anchor and the loop adds μ·w·L to it, which on a long run can exceed the leg force several times; enter μ and the weight per metre (with the length on slides) to include it.

## 8. Verification

Template bs.pipe-expansion@1.3.0 Content hash 8dd3b0517fce49e5 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

## 9. References

1. ASME B31.3.
2. EN 13480-3.